ARTHROPLASTY · NEET-SS / FET / INI-SS

Unicompartmental and osteotomy

1. The spectrum for knee osteoarthritis

- **Non-operative first:** weight loss, quadriceps strengthening and exercise (strong evidence), analgesia (paracetamol, topical/oral NSAIDs), intra-articular steroid (short-term), bracing (unloader), walking aid in the contralateral hand.
- **Arthroscopic debridement/lavage** for degenerative knees: **no benefit** (Moseley 2002, Kirkley 2008); **arthroscopic partial meniscectomy for degenerative tears** is no better than physiotherapy or sham (**FIDELITY**).
- **Joint-preserving surgery** (osteotomy) for **young, active patients with unicompartmental disease and malalignment**; **unicompartmental knee arthroplasty (UKA)** for isolated compartment disease in suitable patients; **TKA** for tricompartmental or advanced disease.

2. High tibial osteotomy (HTO)

Rationale

- **Varus knee + medial compartment OA**: shift the **mechanical axis (weight-bearing line)** from the diseased medial compartment to the **lateral compartment**: the target is the **Fujisawa point**, about **62-66% of the tibial plateau width** from medial (**30-40% beyond the midpoint, i.e. slight overcorrection into about 3-5 degrees of mechanical valgus**).

Patient selection (ideal)

- **Age under about 55-60**, active/heavy labour, **BMI under about 30**.
- **Isolated medial compartment OA** (Ahlback grades I-II; Kellgren-Lawrence under 4), **intact lateral compartment and**

patellofemoral joint (mild PFJ changes acceptable).

- **Range of motion: flexion over about 90-100 degrees**, **fixed flexion deformity under about 10-15 degrees**.
- **Varus deformity under about 10-15 degrees**, **metaphyseal varus (tibial origin: medial proximal tibial angle (MPTA) under 85 degrees)**: the deformity must be in the tibia to correct in the tibia.
- **Stable knee** (ACL deficiency can be addressed with slope change: reduce slope for ACL-deficient knees; increase slope for PCL-deficient knees).
- **Contraindications:** inflammatory arthritis, **lateral compartment disease or meniscectomy**, **tibial subluxation over 1 cm**, severe PFJ OA, poor ROM, smokers (non-union).

Planning

- **Long-leg standing radiographs (hip-to-ankle):** mechanical axis, **Miniaci method** (draw the desired weight-bearing line through the Fujisawa point, rotate around the hinge point to calculate the correction angle/wedge), **joint line convergence angle (JLCA, soft-tissue laxity: overcorrection risk if JLCA over 2-3 degrees)**. Ensure post-op **MPTA under about 95 degrees** (avoid excessive joint line obliquity: shear on cartilage) and decide whether a **double-level osteotomy** (femur + tibia) is needed for combined deformity.

Techniques

	Medial opening-wedge HTO	**Lateral closing-wedge HTO (Coventry)**
Approach	Medial; osteotomy from medial towards the **lateral hinge** (about 1 cm from the lateral cortex, towards the tip of the fibular head)	Lateral; wedge removed; **fibular osteotomy or proximal tibiofibular joint disruption** needed

	Medial opening-wedge HTO	Lateral closing-wedge HTO (Coventry)
Advantages	**No fibular osteotomy**, **no peroneal nerve risk**, adjustable correction intraoperatively, preserves bone stock (easier later TKA), can address the **sagittal plane**	Bone-on-bone contact: faster union, early weight bearing, no graft
Disadvantages	**Increases posterior tibial slope** (if the gap is not made trapezoidal: gap posteromedially about twice anteromedially), **patella baja** (biplanar osteotomy behind the tubercle avoids it), **graft/bone substitute** for larger corrections, **delayed union**, **lateral hinge fracture** (Takeuchi types I-III: type II/III unstable: loss of correction)	**Peroneal nerve injury**, **leg shortening**, **decreases slope**, **patella baja (less)**, **lateral tibial bone loss** complicates later TKA (lateral offset, difficult exposure)
Fixation	**Angle-stable locking plate (TomoFix)**	Staples/plate

- **Outcomes:** about **75-90% survival at 10 years** (conversion to TKA as endpoint); worse with older age, higher BMI, advanced OA, undercorrection (**undercorrection is the commonest cause of failure**).
- **Complications:** under/overcorrection, loss of correction, non-union (opening-wedge, smokers), **lateral hinge fracture**, **peroneal nerve palsy (closing wedge)**, **popliteal artery injury** (protect posteriorly with a retractor, keep the knee flexed), infection, compartment syndrome, DVT, **change in posterior tibial slope**, patella baja.

3. Distal femoral osteotomy (DFO)

- **Valgus knee + lateral compartment OA**, **femoral-origin valgus (lateral distal femoral angle (LDFA) under about 85-87 degrees)**: **varus-producing DFO**.
- **Medial closing-wedge** (stable, faster union) or **lateral opening-wedge** (easier, graft needed; **IT band irritation** from plate). Target: mechanical axis to about **50% (neutral)** or slightly medial (**do not overcorrect into varus**: unlike HTO).
- Also used to unload **lateral osteochondral grafts/meniscal transplant**, correct **patellofemoral malalignment**.
- **Valgus knee correction in the tibia** is avoided because it creates **excessive joint line obliquity** (valgus is usually femoral).

4. Other osteotomies around the knee (brief)

- **Tibial tubercle osteotomy (Fulkerson anteromedialisation)** for patellofemoral malalignment/lateral facet arthritis (see sports notes).
- **Slope-changing (anterior closing-wedge) osteotomy** for revision ACL with **posterior tibial slope over about 12 degrees**.
- **Double-level osteotomy** for severe deformity in both bones (to keep the joint line horizontal).
- **Opening-wedge HTO + ACL reconstruction** for varus ACL-deficient knees (**double/triple varus**: Noyes classification of varus: primary (bony varus), double (+ lateral tibiofemoral separation/LCL laxity), triple (+ posterolateral corner insufficiency, varus recurvatum)).

5. Unicompartmental knee arthroplasty (UKA)

Classical indications (Kozinn and Scott, 1989) vs the Oxford criteria

- **Kozinn and Scott:** age **over 60**, **weight under 82 kg**, low demand, **flexion over 90 degrees**, **fixed flexion under 5 degrees**, **angular**

deformity under 15 degrees (correctable passively), **no inflammatory arthritis**, **no chondrocalcinosis**, intact ACL, **no PFJ exposed bone**.

- **Oxford group (Goodfellow, Murray) broadened: anteromedial OA** (the classic disease pattern: **bone-on-bone medial with full-thickness cartilage preserved laterally and posteriorly in the medial compartment**, **functionally intact ACL**, **correctable varus**, **intact MCL**), **regardless of age, weight, activity, chondrocalcinosis and PFJ status (unless lateral PFJ bone loss with grooving/subluxation)**. **Spontaneous osteonecrosis (SONK/SIFK)** of the medial condyle is also an indication.
- **Contraindications:** inflammatory arthritis, **ACL deficiency** (for mobile-bearing: fixed-bearing UKA ± ACL reconstruction in young patients), **lateral compartment disease** (for medial UKA), **fixed deformity not correctable**, previous HTO (relative), **lateral PFJ damage with bone loss**.
- **Pre-op workup: valgus stress radiograph** (lateral compartment full-thickness cartilage preserved: lateral joint space maintained) and **varus stress radiograph** (bone-on-bone medially: confirms full-thickness loss; **avoid UKA for partial-thickness disease**: high rate of unexplained pain/revision).

Designs

	Mobile-bearing (Oxford)	**Fixed-bearing (metal-backed or all-polyethylene)**
Conformity	**Fully congruent spherical femur and bearing**: low contact stress, **minimal polyethylene wear**	Less conformity: higher contact stress

	Mobile-bearing (Oxford)	Fixed-bearing (metal-backed or all-polyethylene)
Main issue	**Bearing dislocation** (especially **lateral UKA: about 10%** mobile-bearing; use fixed-bearing laterally or a domed lateral design)	Polyethylene wear, loosening
Requirements	**Intact, functioning ligaments** (ACL, MCL)	Slightly more forgiving of ligaments

- **Robotic-assisted UKA** improves component positioning accuracy.

Principles of surgery

- **Do not overcorrect** the deformity: **correct to the pre-disease alignment**, not neutral (**overcorrection overloads the opposite compartment: progression of OA there**); leave the **MCL unreleased** (no soft-tissue release in medial UKA).
- **Minimally invasive approach**; preserve the ACL.

Outcomes and failures

- **Advantages over TKA:** faster recovery, **more natural kinematics and better ROM**, lower morbidity (**lower mortality, VTE, MI, infection, transfusion** in registry analyses), better function ("forgotten knee"), less blood loss, shorter stay.
- **Disadvantages: higher revision rate in registries** (about **2-3 times that of TKA**), partly due to the low threshold for revision (a UKA is easier to revise than a TKA), surgeon volume (**surgeons doing over about 20% of their knee arthroplasties as UKA**, or 13+ per year, have better results); **TOPKAT trial** (UKA vs TKA in medial OA): **similar outcomes and revision at 5 years**, UKA more cost-effective.
- **Causes of failure: progression of OA in the other compartments** (lateral), **aseptic loosening (tibial)**, **unexplained pain**, **bearing dislocation** (mobile), polyethylene wear (fixed), **tibial plateau**

fracture (usually early: excessive vertical saw cut, pin holes), infection.

- **Radiolucent lines** under the tibial component of the Oxford UKA (**physiological, thin, well-defined, under 1-2 mm, non-progressive**) are **common and do not indicate loosening**: **do not revise** for them alone.
- **Lateral UKA:** less common (**about 5-10% of UKAs**); lateral compartment has more **rollback and laxity** in flexion; valgus deformity often has a femoral component.

6. Patellofemoral arthroplasty (PFA)

- **Isolated patellofemoral OA** (often **trochlear dysplasia**-related) in younger patients; **onlay** trochlear designs have fewer patellar maltracking problems than **inlay** designs.
- **Failure: progression of tibiofemoral OA** (commonest), maltracking, pain; convert to TKA.

7. Bicompartmental and other options

- **Bi-unicompartmental** and **bicompartmental (medial UKA + PFA)** arthroplasty: niche.
- **Cartilage procedures** for focal defects (see sports notes) are distinct from OA.

8. HTO vs UKA in medial OA (choosing)

Feature	Favours HTO	Favours UKA
Age	Younger (**under 55-60**)	Older (**over 60**) (Oxford: any age)
Activity	**High (manual labour, sports)**	Lower to moderate
Deformity	**Tibial (metaphyseal) varus, constitutional**	Intra-articular varus from cartilage loss (correctable)

Feature	Favours HTO	Favours UKA
OA severity	**Mild-moderate (partial thickness)**	**Bone-on-bone (full-thickness)**, SONK
ACL	Can be deficient (combine with ACL reconstruction)	**Must be intact** (mobile-bearing)
ROM	Good (flexion over 90)	Good

Exam pearl: **HTO: young, active, isolated medial OA, tibial varus (MPTA under 85); target the Fujisawa point (62-66% of plateau width). Opening-wedge increases slope and causes patella baja; closing-wedge risks the peroneal nerve. Commonest cause of HTO failure: undercorrection.**

Exam pearl: **DFO for valgus knee with femoral deformity: correct to neutral, not overcorrected. Valgus is corrected on the femoral side to avoid joint line obliquity.**

Exam pearl: **Oxford UKA: anteromedial OA with full-thickness medial loss, functioning ACL, correctable varus, intact lateral compartment (valgus stress view). Do not overcorrect; do not release the MCL. Physiological tibial radiolucent lines are not loosening. Mobile-bearing lateral UKA = bearing dislocation. UKA: higher registry revision but lower morbidity than TKA (TOPKAT similar).**

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.